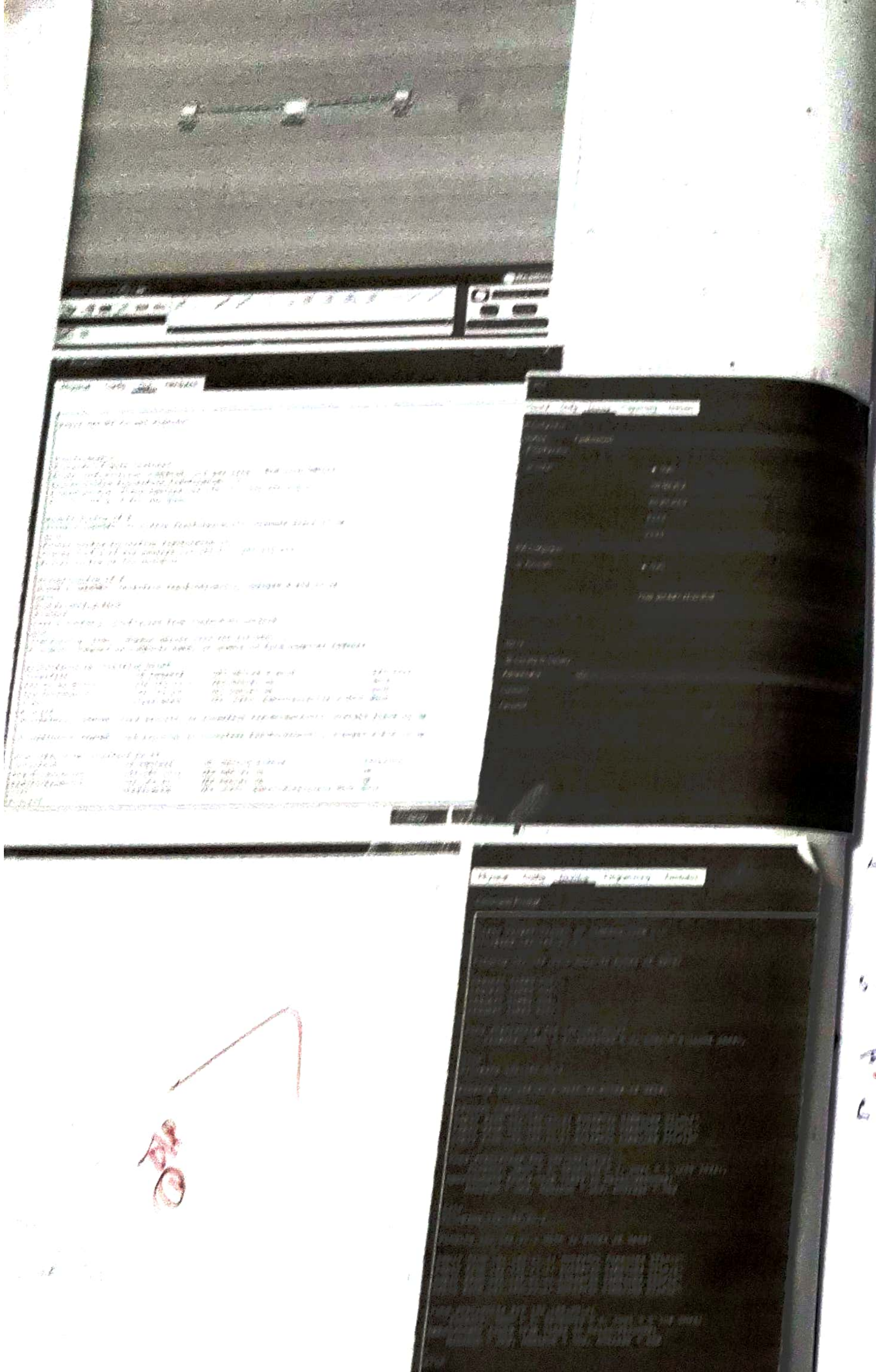


Exp: 10. (a)
Date: 18/9/26

Internetworking with Routers
in Cisco packet Tracer

(a)
Design and configure a simple Internetwork using a Router

Aim:

To design and configure an internetworks consisting of multiple network connected through a router and observe packet transmission b/w different network.

Algorithm /procedure

1) open cisco Packet Tracer.

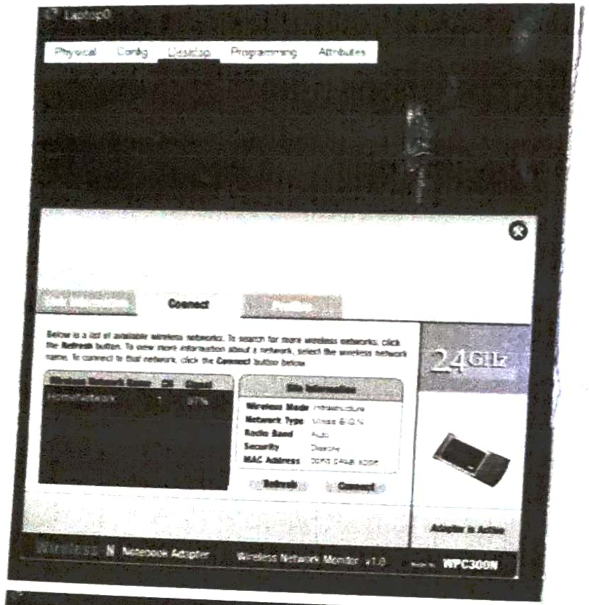
2) place a router switches using copper straight through cables

3) connect the PC to switches using copper straight through cables

4) connect each switch to separate router interface.

5) Enable the router interface using the no ~~shut down~~ command.

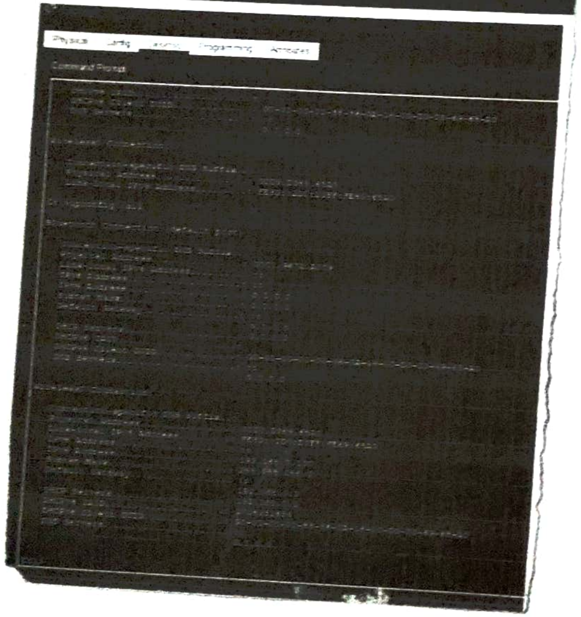
6. Configure the default gateway of each PC using the IP address of its local Router interface.



Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.

HomeNetwork 1 87%

Wireless Mode Infrastructure
Network Type Mixed B/G/N
Radio Band Auto
Security Disable

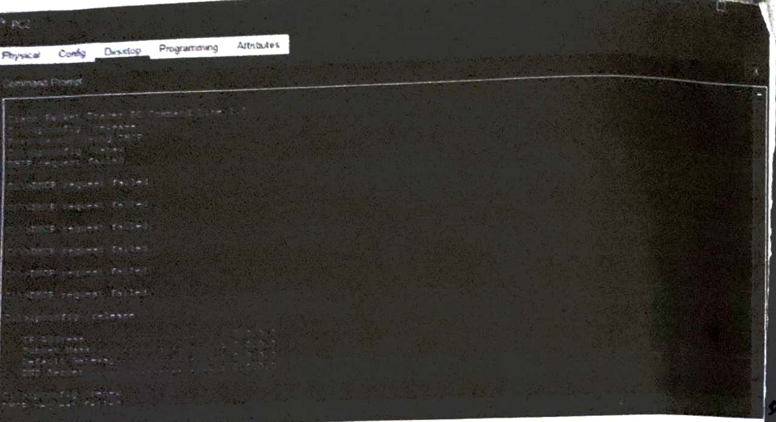


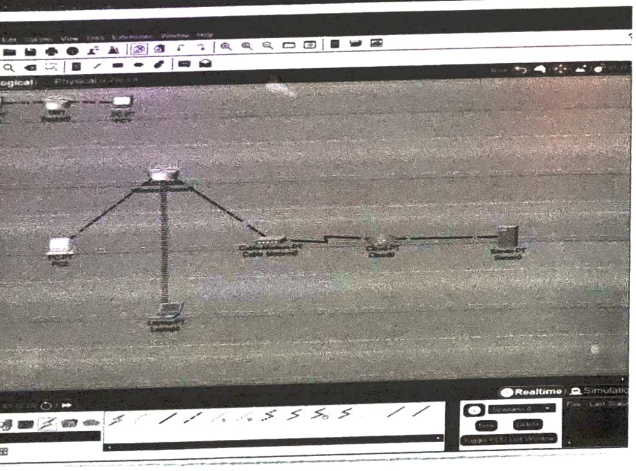
4) Test connectivity b/w pcs in different network using the ping command.

5) Use simulation mode to observe flow packets through the router

:

Result: A simple internetwork was designed and configured successfuly communication b/w different networks was verified and packet forwarding through the router was observed.





## b) Internetwork using wireless Router DHCP server and Internet cloud.

AIM:- To configure an internetwork using a wireless router DHCP server and Internet cloud in cisco packet tracer.

Algorithm / procedure

1) open cisco packet tracer and place wireless router, wireless and devices a server and. and Internet cloud.

2. configure the wireless router with an SSID and security settings

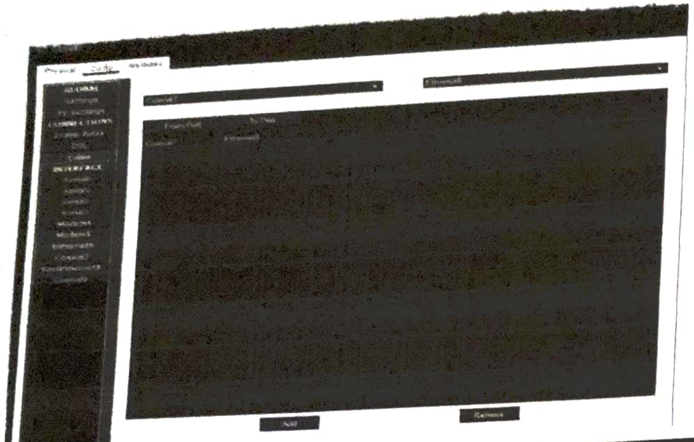
3) Enable DHCP and with a suitable static address and enable the required services.

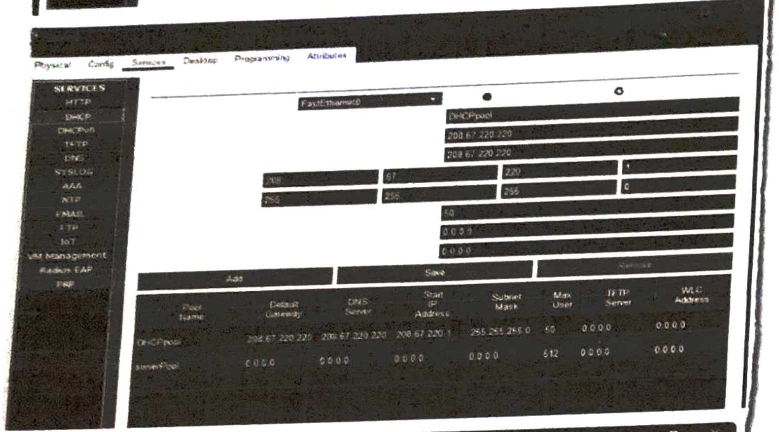
4) connect the router to the Internet cloud using the ~~p~~ appropriate interface and connection

5) connect wireless devices to the configured. the appropriate interface and connection

6) connect wireless devices to the configured SSID

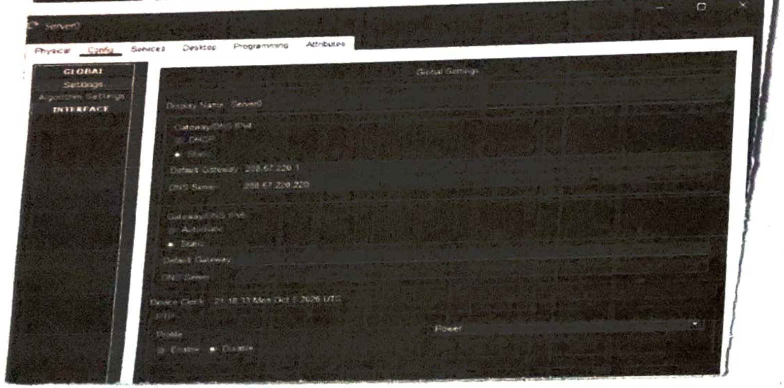
7) verify that the devices receive IP address automatically through DHCP.





DHCPpool
208.67.220.220
208.67.220.220
208 | 67 | 220 | 1
255 | 255 | 255 | 0
50
0.0.0.0
0.0.0.0

| Pool Name | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|---|---|---|---|---|---|---|---|
| DHCPpool | 208.67.220.220 | 208.67.220.220 | 208.67.220.1 | 255.255.255.0 | 50 | 0.0.0.0 | 0.0.0.0 |
| serverPool | 0.0.0.0 | 0.0.0.0 | 0.0.0.0 | 0.0.0.0 | 512 | 0.0.0.0 | 0.0.0.0 |



Display Name: Server0

Gateway/DNS IPv4
○ DHCP
● Static
Default Gateway: 208.67.220.1
DNS Server: 208.67.220.220

Gateway/DNS IPv6
○ Automatic
● Static
Default Gateway
DNS Server

Device Clock: 21:18:33 Mon Oct 5 2026 UTC
PTP
Profile: Power
○ Enable ● Disable

8) Test connectivity to the server and other reachable network devices.

Results:

The wireless network was configured successfully. Devices obtained IP address automatically through DHCP and connectivity was tested.